

Intro

Hazel Kim

Research Interest

Natural Language Processing
Machine Reasoning

Future Goals

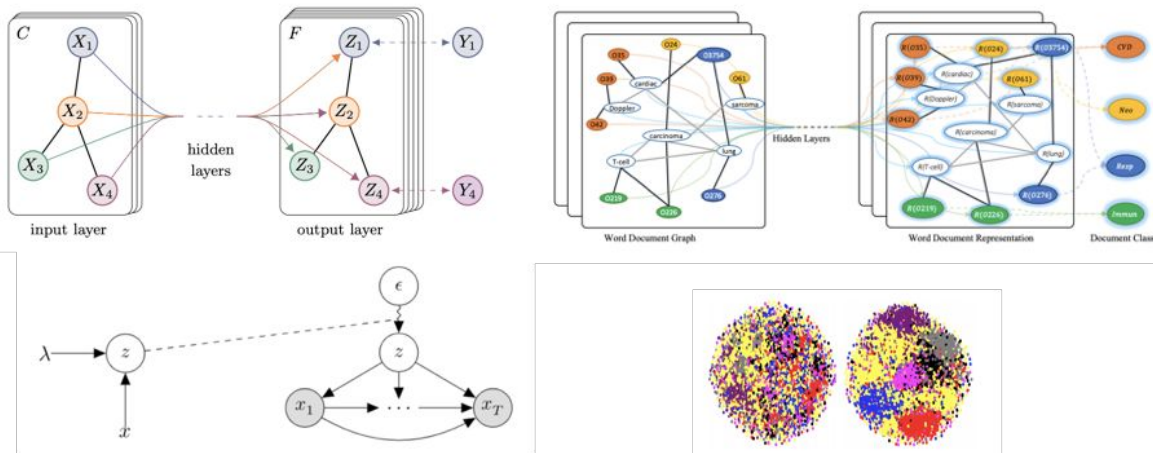
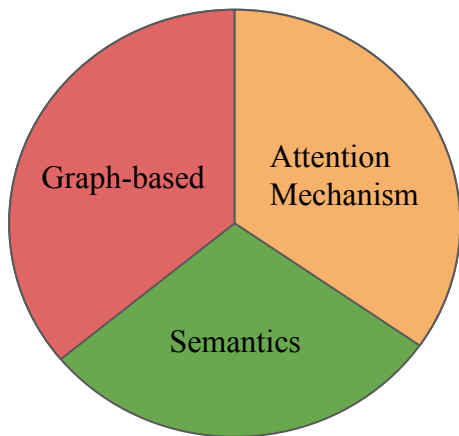
- Level of Artificial Intelligence that I want to reach:
 - Among the social sciences, **law** may come the closest to **a system of formal logic**. To oversimplify, legal rulings involve setting forth axioms derived from precedent, applying those axioms to the particular facts at hand, and reaching a conclusion accordingly. This logic-oriented methodology is exactly the type of activity to which machine intelligence can fruitfully be applied.
 - AI Will Transform The Field Of Law - Fobes 12/19/2019
 - Contract Review
 - Contract Analytics
 - Litigation Prediction
 - Legal Research
 - “AI technology will ultimately broaden the lawyer’s role from a narrow focus on risk mitigation to more strategic engagement on company initiatives.” - Lawgeex CEO Noory Bechor.

Future Goals

- Areas that I want to get involved:
 - Latent Alignment for Semantic Analysis
 - Interpretable and Generalizable Deep Learning
 - Logic
 - Linguistic Perspective + Cognitive Science
 - Language serves essentially for the expression of thought. - Noam Chomsky
 - Language is the jewel in the crown of cognition. - Steven Pinker

Previous Research

- Graph-based (Heterogenous)
- Attention Mechanism
- Semantics



References from <https://arxiv.org/pdf/1812.06834.pdf>, and <https://arxiv.org/pdf/1809.05679.pdf>

- **Preserving semantic knowledge** for better text analysis: Use of Graph's Topological Structures.
- **Latent Semantic Analysis:** An NLP Technique for conceptual analysis which produces concepts to analyze relationship between a set of documents and the terms they contain.

Previous Research

1. Gated Attention Graph Network
2. Graph Bert for Text Classification
3. Distributional Semantics

Previous Research 1: Gated Attention Graph Network

Research Objectives

- Convolutional Operator + Attention Mechanism + Different Importance
 - Graph Aggregator

Previous Research 1: Gated Attention Graph Network

Data Exploration

Node Features

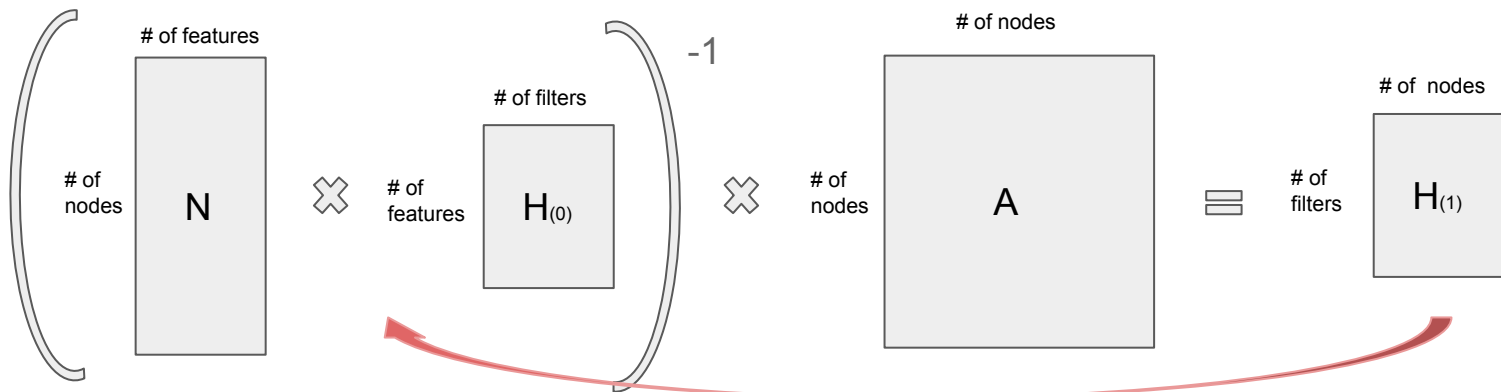
- Row: node size = 12772
- Column: feature categories = 1256

Edges (size = 12772)

- Movie-Director: 4661 stored elements
- Director-Movie: 4661 stored elements
- Actor-Movie: 13983 stored elements
- Movie-Actor: 13983 stored elements

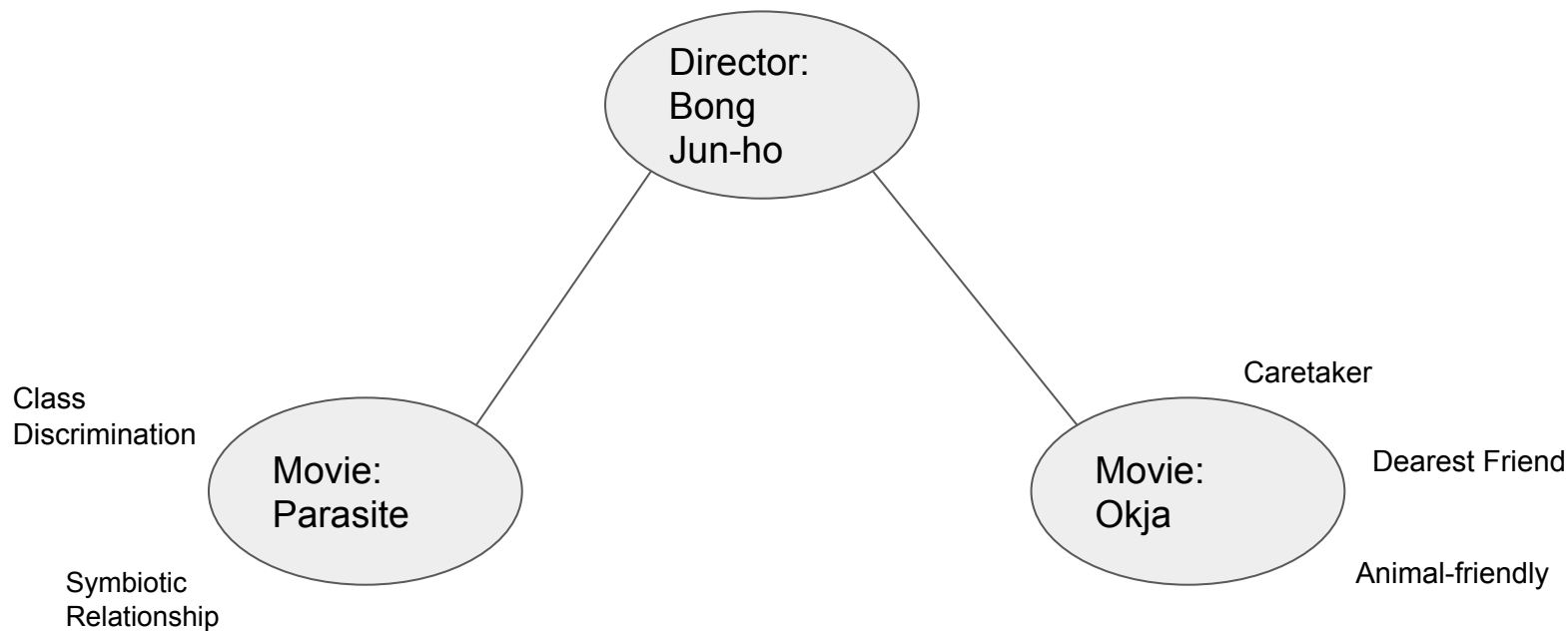
Labels (0: drama, 1: comedy, 2: action)

- Train: 300
- Validation: 300
- Test: 2339



Previous Research 1: Gated Attention Graph Network

Node Features



Previous Research 1: Gated Attention Graph Network

Evolution of GNN

- Graph Convolutional Network (GCN), 2017
 - Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations (ICLR)*, 2017.
- GraphSAGE, 2017
 - W. Hamilton, Z. Ying, and J. Leskovec. Inductive representation learning on large graphs. In *NIPS*, pages 1025–1035, 2017a.
- Graph Attention Network (GAT), 2018
 - P. Velićković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio. Graph attention networks. In *ICLR*, 2018.
- Gated Attention Network (GaAN), 2018
 - Zhang, Jiani, et al. "Gaan: Gated attention networks for learning on large and spatiotemporal graphs." *arXiv preprint arXiv:1803.07294* (2018).

Previous Research 1: Gated Attention Graph Network

Graph Aggregator

Develops a localized parameter-sharing operator on a set of neighboring nodes to aggregate a local set of lower-level features. (Using a receptive field of the aggregator)

- Its design determines a model's ability to capture the structural information of graphs.
- Functions that are permutation invariant and can be dynamically resizing
- i.e. either pooling over neighborhoods or computing a weighted sum of the neighboring features.

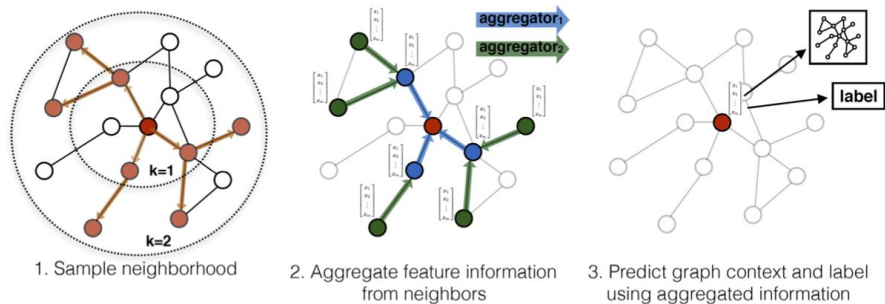


Figure 1: Visual illustration of the GraphSAGE sample and aggregate approach.

Previous Research 1: Gated Attention Graph Network

How to Learn Representations of Nodes in a Graph

Algorithm 1: WL-1 algorithm (Weisfeiler & Lehmann, 1968)

Input: Initial node coloring $(h_1^{(0)}, h_2^{(0)}, \dots, h_N^{(0)})$

Output: Final node coloring $(h_1^{(T)}, h_2^{(T)}, \dots, h_N^{(T)})$

$t \leftarrow 0$;

repeat

for $v_i \in \mathcal{V}$ **do**

$h_i^{(t+1)} \leftarrow \text{hash} \left(\sum_{j \in \mathcal{N}_i} h_j^{(t)} \right)$;

$t \leftarrow t + 1$;

until *stable node coloring is reached*;

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}_i} \frac{1}{c_{ij}} h_j^{(l)} W^{(l)} \right)$$

- c_{ij} is an appropriately chosen normalization constant for the edge (v_i, v_j)
- $h^{(l)}$ now to be a vector of *activations* of node i in the l th neural network layer
- $W^{(l)}$ is a (l) layer-specific weight matrix

Replace the hash function with a neural network layer-like differentiable function with trainable parameters.

Previous Research 1: Gated Attention Graph Network

Different Graph Aggregators

GCN

Scales linearly in the number of graph edges.

Learns hidden layer representations that encode both local graph structure and features of nodes.

GraphSAGE

(SAmple + AggreGatE)

Compute only the representations that are necessary to satisfy the recursion at each depth.

Operates by sampling a fixed-size neighborhood of each node, and then perform a specific aggregator over it.

- Mean Aggregator
- Pooling Aggregator

GAT

Uses a sampling algorithm to select a small subset of the nodes and edges.

To obtain sufficiently expressive power to transform the input features into higher level, the model adds learnable linear transformation (attention mechanism).

Does not compute additional value vectors and uses a fully-connected layer to compute ϕ_w .

GaAN

Adopts the key-value attention mechanism and the dot product attention.

Uses convolutional network that takes the center node and neighboring node features to generate the gate values.

Adding gates would not produce too many additional parameter. (Computationally Inexpensive)

Previous Research 1: Gated Attention Graph Network

GaAN

- G vector: **combination of average pooling and max pooling.**

$$\mathbf{y}_i = \text{FC}_{\theta_o}(\mathbf{x}_i \oplus \bigoplus_{k=1}^K (g_i^{(k)} \sum_{j \in \mathcal{N}_i} w_{i,j}^{(k)} \text{FC}_{\theta_v^{(k)}}(\mathbf{z}_j))),$$

$$\mathbf{g}_i = [g_i^{(1)}, \dots, g_i^{(K)}] = \psi_g(\mathbf{x}_i, \mathbf{z}_{\mathcal{N}_i}),$$

$$\mathbf{g}_i = \text{FC}_{\theta_g}^\sigma(\mathbf{x}_i \oplus \max_{j \in \mathcal{N}_i} \{\text{FC}_{\theta_m}(\mathbf{z}_j)\}) \oplus \frac{\sum_{j \in \mathcal{N}_i} \mathbf{z}_j}{|\mathcal{N}_i|})$$

- θ_m maps the neighbor features to a d_m dimensional vector before taking the element-wise max.
- θ_g maps the concatenated features to the final K gates.
- By setting a small d_m , the subnetwork for computing the gate will have negligible computational overhead.

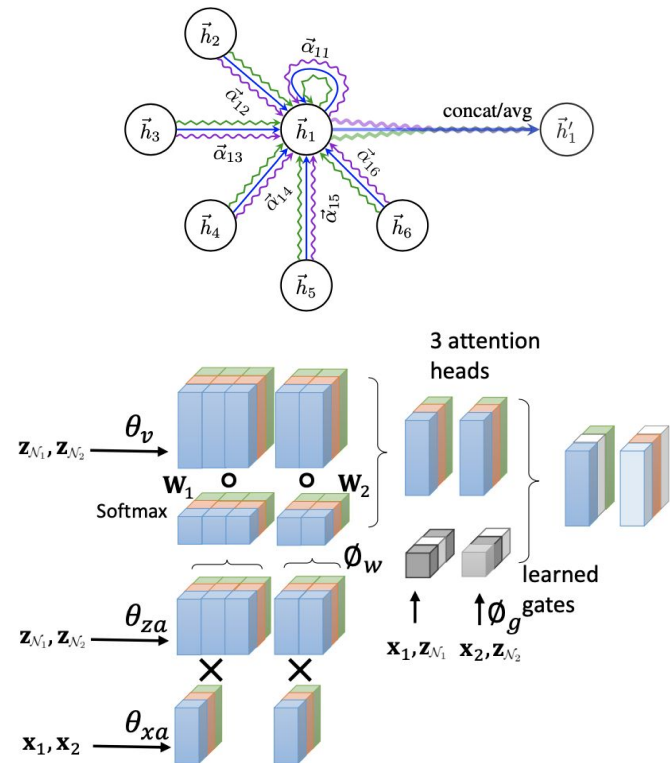


Figure 1: Illustration of a three-head gated attention aggregator with two center nodes in a mini-batch. $|\mathcal{N}_1| = 3$ and $|\mathcal{N}_2| = 2$ respectively. Different colors indicate different attention heads. Gates in darker color stands for larger values. (Best viewed in color)

Previous Research 1: Gated Attention Graph Network

Take Advantage of GaAN

- Sampling a small number of neighborhood size instead of considering all.
- Various attention alignment available.

Table 4: Comparison of the test F1 score on the Reddit and PPI datasets with different sampling neighborhood sizes and attention head number K . S_1 and S_2 are the maximum number of sampled neighborhoods in the 1st and 2nd sampling steps. ‘all’ means to sample all the neighborhoods.

Models	#Param	Reddit				PPI	
		S_1, S_2 25,10	S_1, S_2 50,20	S_1, S_2 100,40	S_1, S_2 200,80	#Param	S_1, S_2 all, all
2-Layer FNN	1.71M	73.58±0.09	73.58±0.09	73.58±0.09	73.58±0.09	1.23M	54.07±0.06
Avg. pooling	866K	95.78±0.07	96.11±0.07	96.28±0.05	96.35±0.02	274K	96.85±0.19
Max pooling	866K	95.62±0.03	96.06±0.09	96.18±0.11	96.33±0.04	274K	98.39±0.05
Pairwise+sigmoid	965K	95.86±0.08	96.19±0.04	96.33±0.05	96.38±0.08	349K	98.39±0.05
Pairwise+tanh	965K	95.80±0.03	96.11±0.05	96.26±0.03	96.36±0.04	349K	98.32±0.18
Attention-only-K1	562K	96.15±0.06	96.40±0.05	96.48±0.02	96.54±0.07	168K	96.31±0.08
Attention-only-K2	571K	96.19±0.07	96.40±0.04	96.52±0.02	96.57±0.02	178K	97.36±0.08
Attention-only-K4	587K	96.11±0.06	96.40±0.02	96.49±0.03	96.56±0.02	196K	98.09±0.07
Attention-only-K8	620K	96.10±0.03	96.38±0.01	96.50±0.04	96.53±0.02	233K	98.46±0.09
GaAN-K1	620K	96.29±0.05	96.50±0.08	96.67±0.04	96.73±0.05	201K	96.95±0.09
GaAN-K2	629K	96.33±0.02	96.59±0.02	96.71±0.05	96.82±0.05	211K	97.92±0.05
GaAN-K4	645K	96.36±0.03	96.60±0.03	96.73±0.04	96.83±0.03	230K	98.42±0.02
GaAN-K8	678K	96.31±0.13	96.60±0.02	96.75±0.03	96.79±0.08	267K	98.71±0.02

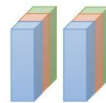
Previous Research 1: Gated Attention Graph Network

Implementation: Gated Attention Aggregator

- Class Attention(nn.Module)

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\vec{\mathbf{a}}^T[\mathbf{W}\vec{h}_i \parallel \mathbf{W}\vec{h}_j]\right)\right)}{\sum_{k \in \mathcal{N}_i} \exp\left(\text{LeakyReLU}\left(\vec{\mathbf{a}}^T[\mathbf{W}\vec{h}_i \parallel \mathbf{W}\vec{h}_k]\right)\right)}$$

3 attention heads



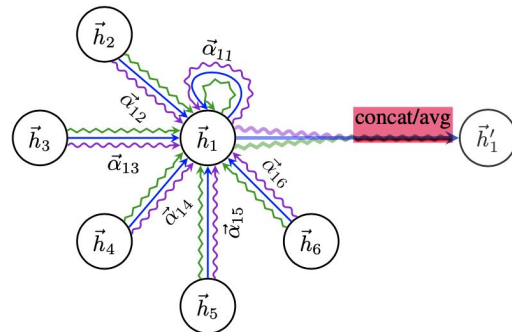
- Class GCN Layer: forward()
- Class GCN Block: layers.append()
- Class GCN Net: forward()

$$\mathbf{y}_i = \text{FC}_{\theta_o}(\mathbf{x}_i \oplus \bigoplus_{k=1}^K (g_i^{(k)} \sum_{j \in \mathcal{N}_i} w_{i,j}^{(k)} \text{FC}_{\theta_v^{(k)}}(\mathbf{z}_j))),$$

$$\mathbf{g}_i = [g_i^{(1)}, \dots, g_i^{(K)}] = \psi_g(\mathbf{x}_i, \mathbf{z}_{\mathcal{N}_i}),$$

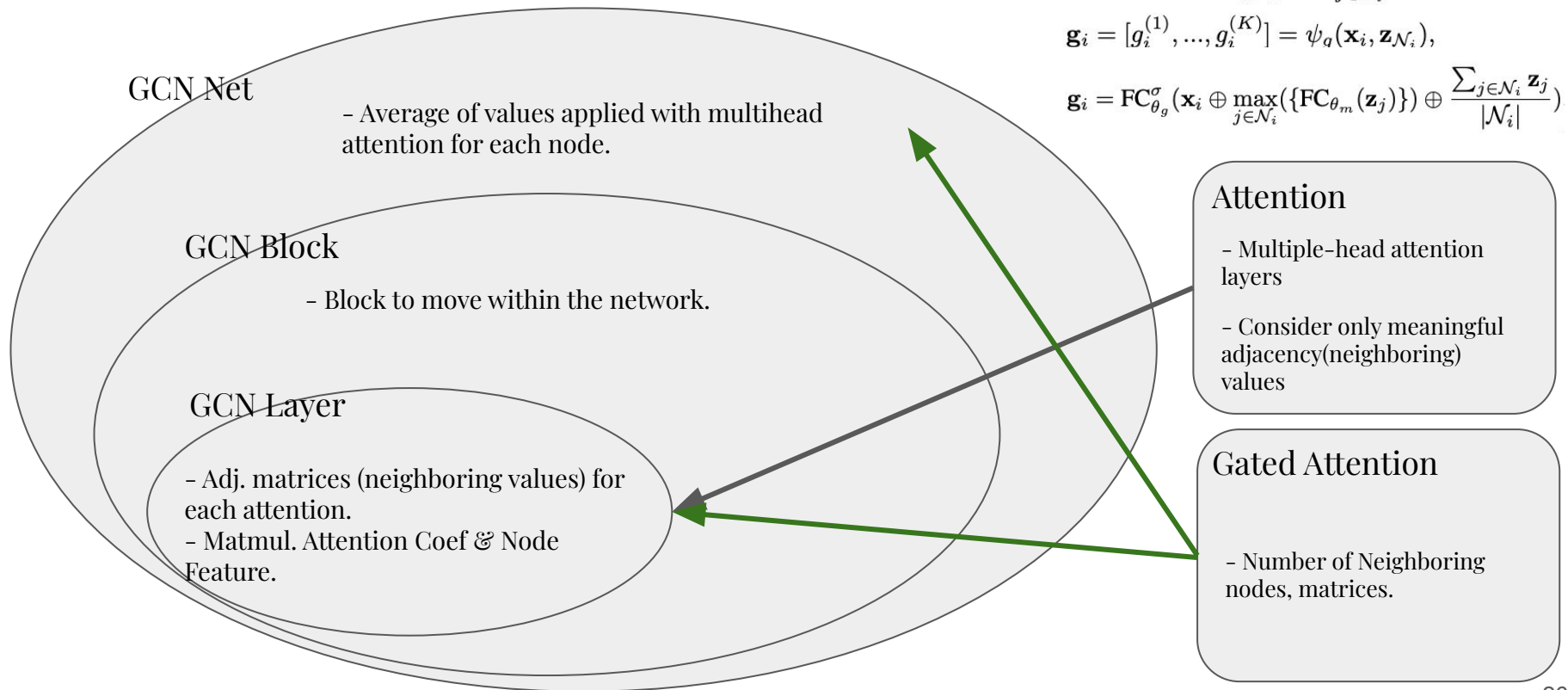
g vector operation

- `out = torch.sum(out, dim=0)`
- `out = torch.div(out, num_head)`
- `out = self.elu(out)`



Previous Research 1: Gated Attention Graph Network

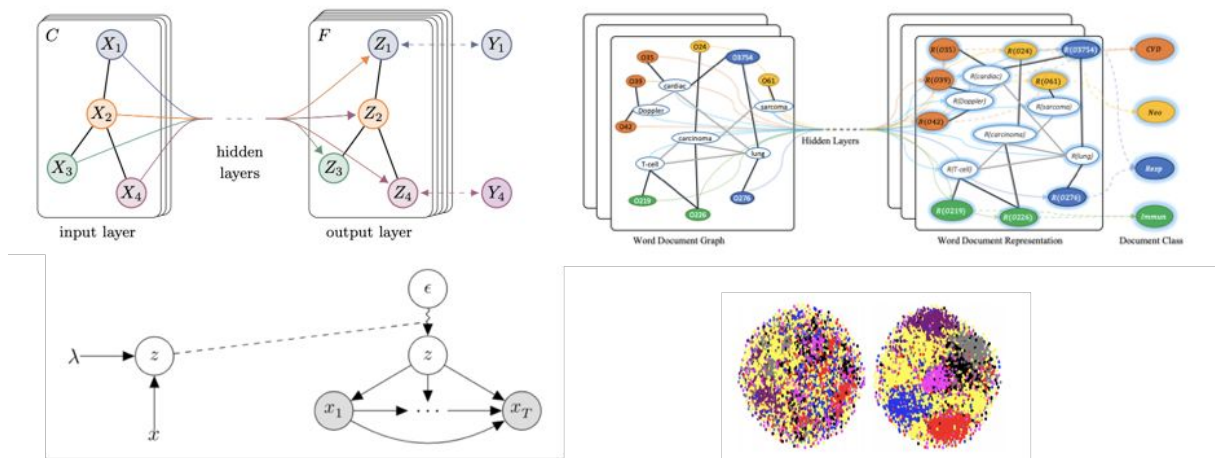
Network Implementation Structure



Previous Research 1: Gated Attention Graph Network

Challenges

- **Preserving semantic knowledge for better text analysis:** Use of Graph's Topological Structures.
- **Latent Semantic Analysis:** An NLP Technique for conceptual analysis which produces concepts to analyze relationship between a set of documents and the terms they contain.
- **Yet, lack of explainability for the latent space.**



Previous Research 3: Distributional Semantics

Proposal: Distributional Semantics, why important?

- Individual Corpora
 - “Meaning is negotiated between two people.”
 - Words from every speaker with every possible use are not actually realistic in terms of what an individual understands a meaning of a word.
 - We crucially don’t know about negotiation of meaning between speakers when we get a very large text corpus.
 - Corpora which are individuated so that we know what an individual speaker hears and says as opposed to a very large, huge text corpora we have. (Variations between speakers)
- A longitude of Corpora
 - After serious scientific investigation on individual corpora.
 - Corpora collected from individual speakers help work on syntax, speaker variation.
- Proposal by Prof. Copestake
 - In 2013 Berlin, <https://www.youtube.com/watch?v=b4oUp7KPseA>

Previous Research 3: Distributional Semantics

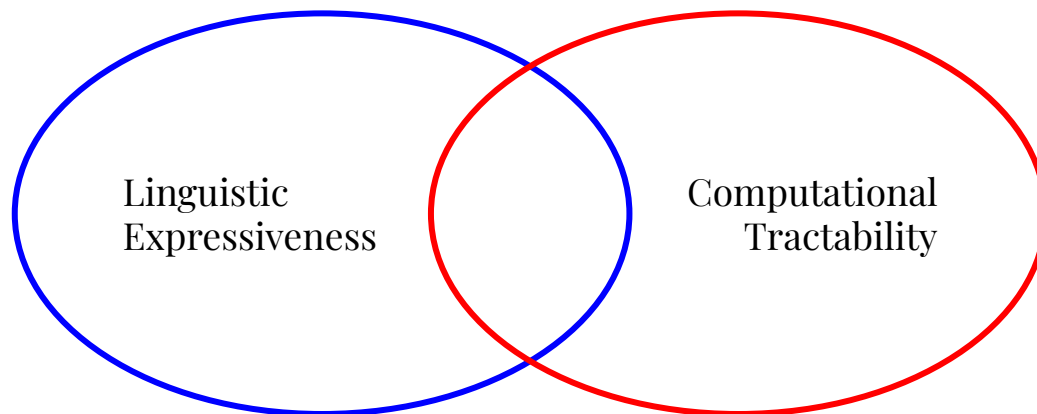
Their Publications

- Guy Emerson and Ann Copestake. 2017. "**Semantic Composition** via Probabilistic Model Theory". In Proceedings of the 12th International Conference on Computational Semantics (IWCS).
- Guy Emerson and Ann Copestake. 2017. "**Variational Inference** for Logical Inference". In Proceedings of the 2017 Conference on Logic and Machine Learning for Natural Language (LaML).
- Guy Emerson and Ann Copestake. 2016. "Functional **Distributional Semantics**". In Proceedings of the ACL 2016 Workshop on Representation Learning for NLP (RepL4NLP).
- Ann Copestake, Guy Emerson, Michael Wayne Goodman, Matic Horvat, Alexander Kuhnle, and Ewa Muszyńska. 2016. "Resources for Building Applications with **Dependency Minimal Recursion Semantics**". In Proceedings of 10th International Conference on Language Resources and Evaluation (LREC).
- Guy Emerson and Ann Copestake. 2015. "Lacking Integrity: HPSG as a Morphosyntactic Theory". In Proceedings of the 22nd International Conference on Head-Driven Phrase **Structure Grammar** (HPSG).
- Guy Emerson and Ann Copestake. 2015. "Leveraging a **Semantically Annotated Corpus** to Disambiguate Prepositional Phrase Attachment". In Proceedings of the 11th International Conference on Computational Semantics (IWCS).

Previous Research 3: Distributional Semantics

Idea Overview

- While linguistic insights can guide the design of model architectures, future progress will require **balancing** the often conflicting demands of linguistic expressiveness and computational tractability.



Previous Research 3: Distributional Semantics

Meaning and the World

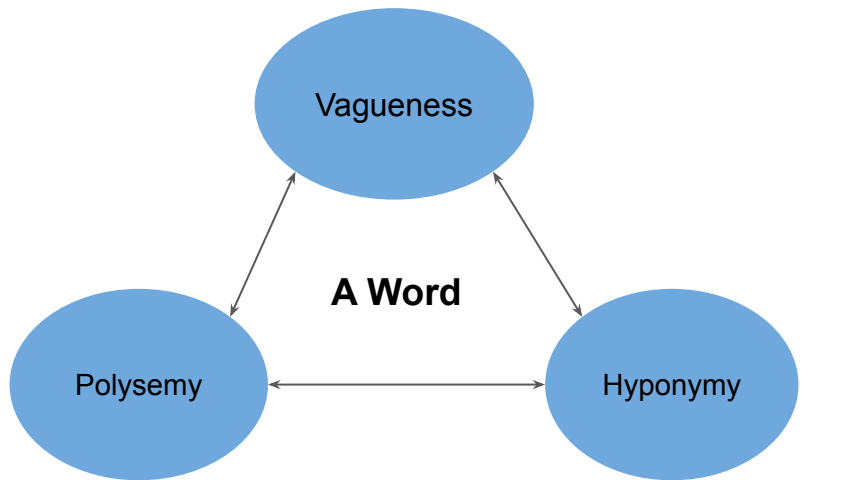
- Grounding
 - Process of connecting language to the world
 - If the meanings of words are defined only in terms of other words, these definitions are circular (Harnad, 1990)
- Approaches
 - Train a distributional model as normal, then combine it with a grounded model.
 - Find correlations between distributional and sensory features.
 - Joint learning
 - Define a single model, whose parameters are learnt based on both corpus data and grounded data.
 - Latent Dirichlet Allocation (LDA) Model
 - Skip-gram model
- Pure distributional models look for word co-occurrence patterns, while joint models prefer co-occurrence patterns that match the grounded data.

Previous Research 3: Distributional Semantics

Semantic Challenges in Linguistic Perspective

- Lexical Meaning

No sharp cutoff between concepts

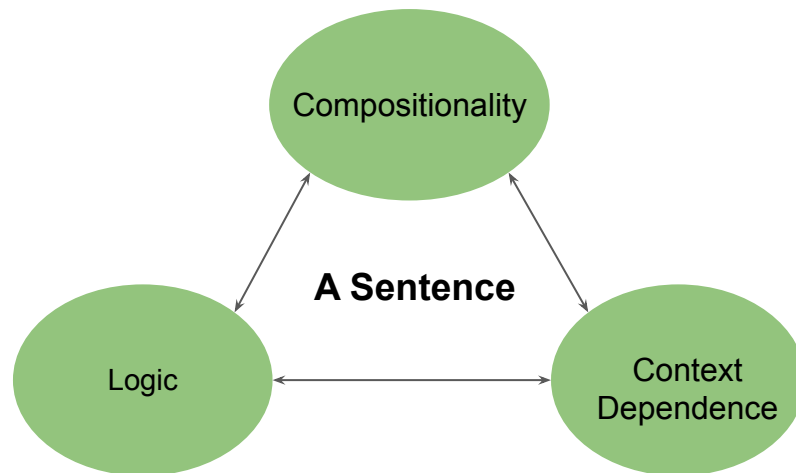


With related senses among distinct

One term that has more general meaning than another

- Sentence Meaning

Ability to derive the meaning of a sentence from its part, and to generalize to new combinations



Complex thoughts and chains of reasoning

The meaning of an expression depending on the context

Previous Research 3: Distributional Semantics

Sentence Meaning: Logic

- A truth-conditional functions that are compatible with first-order logic (Emerson and Copestake, 2017b; Emerson, 2020b)

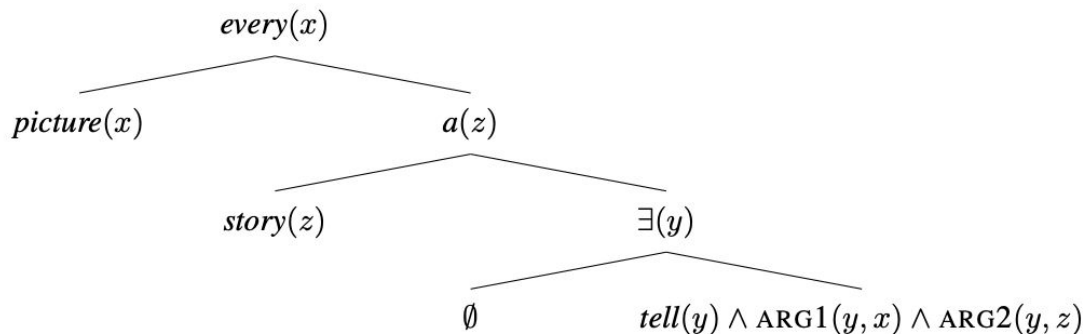
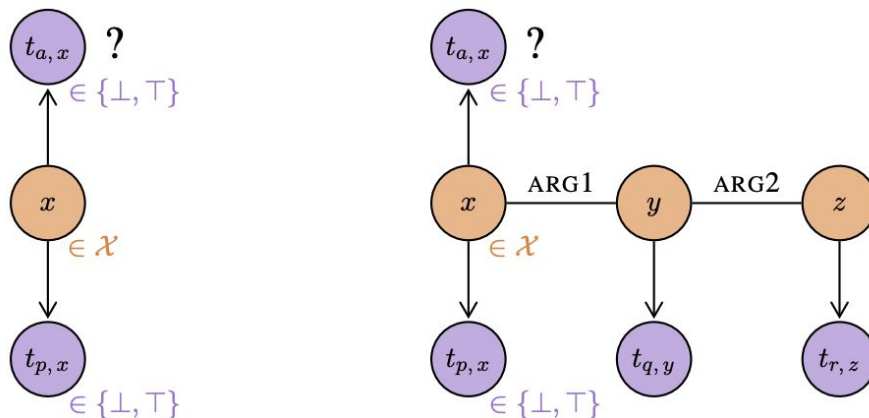


Figure 3: Fully scoped representation of the most likely reading of *Every picture tells a story*. Each non-terminal node is a quantifier, its left child its restriction, and its right child its body. We assume that event variables are existentially quantified with no constraints on the restriction.

Previous Research 3: Distributional Semantics

Sentence Meaning: Context Dependence

- Calculate occasion meanings by conditioning on the context (Emerson and Copestake, 2017b)



(a) Inferring if a is true of x , when there are no further pixies in the situation, and given the truth value for the predicate p .

(b) Inferring if a is true of x , when x is part of a larger situation, and given the truth value for one predicate for each pixie in the situation.

Figure 4: Examples of graphical models for inference. In both cases, we want to find the probability of the predicate a being true of a pixie x , given some other information about the situation.

Previous Research 3: Distributional Semantics

The Aim of Distributional Semantics

- Requires structure beyond representing meaning as a point in a space
 - It seems desirable to represent the meaning of a word as a region of space or as a classifier, and to work with probability logic.
 - However, there is a trade-off between expressiveness and learnability.
 - Promising neural architectures for working with structured data
 - Dependency Graphs
 - Logical Propositions
 - Promising new techniques to mitigate computationally expensive calculations in probabilistic models
 - Amortized variational inference, used in the Variational Autoencoder

References

- Asai, Akari, et al. "Learning to retrieve reasoning paths over wikipedia graph for question answering." *arXiv preprint arXiv:1911.10470* (2019).
- Spinelli, Indro, et al. "Efficient data augmentation using graph imputation neural networks." *arXiv preprint arXiv:1906.08502* (2019).
- Wang, Haoyu, et al. "Do Multi-hop Readers Dream of Reasoning Chains?" *arXiv preprint arXiv:1910.14520* (2019).
- Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding." *arXiv preprint arXiv:1810.04805* (2018).
- Reimers, Nils, and Iryna Gurevych. "Sentence-bert: Sentence embeddings using siamese bert-networks." *arXiv preprint arXiv:1908.10084* (2019).
- Deng, Yuntian, et al. "Latent alignment and variational attention." *Advances in Neural Information Processing Systems*. (2018).
- Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *International Conference on Learning Representations (ICLR)*, 2017.
- W. Hamilton, Z. Ying, and J. Leskovec. Inductive representation learning on large graphs. In *NIPS*, pages 1025–1035, 2017a.
- P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio. Graph attention networks. In *ICLR*, (2018).
- Brendan L. Douglas. The Weisfeiler-Lehman method and graph isomorphism testing. *arXiv preprint arXiv:1101.5211*, 2011.
- Boris Weisfeiler and A. A. Lehmann. A reduction of a graph to a canonical form and an algebra arising during this reduction. *Nauchno-Tekhnicheskaya Informatsia*, 2(9):12–16, (1968).
- Zhou, Jie, et al. "Graph neural networks: A review of methods and applications." *arXiv preprint arXiv:1812.08434* (2018).
- Zhang, Jiani, et al. "Gaan: Gated attention networks for learning on large and spatiotemporal graphs." *arXiv preprint arXiv:1803.07294* (2018).
- Xue, Lanqing, Xiaopeng Li, and Nevin L. Zhang. "Not All Attention Is Needed: Gated Attention Network for Sequence Data." *arXiv preprint arXiv:1912.00349* (2019).
- Emerson, Guy & Copestake, Ann. (2017). Semantic Composition via Probabilistic Model Theory.
- Emerson, Guy. "What are the Goals of Distributional Semantics?." *arXiv preprint arXiv:2005.02982* (2020).